

Assessment Tool -> 02
CO-2 (Scenario Based
Questions).

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Course code: ITA0506.

Course Name: Computer Vision.

Manufacturing Quality Inspection.

(a) Identify the type of noise present.

→ The images contain periodic noise caused by electrical interference. It appears as repetitive horizontal stripes across the image.

(b) Explain how this degradation affects inspection accuracy.

- Hides small defects such as cracks and scratches.

- Produces false edges and patterns.

- Reduces image quality.

- Causes incorrect defect detection.

- Decreases the reliability of the automated inspection system.

(c) Recommend an appropriate enhancement technique.

→ Frequency Domain Filtering, specifically a notch defect filter.

Steps:

1. Apply the Fourier Transform (FFT) to convert the image into the frequency domain.

2. Delete the bright peaks corresponding to periodic noise.

3. Apply a Notch Reject Filter to remove only those frequencies.

4. Perform the Inverse Fourier Transform (IFFT) to obtain the restored image.

(d) Justify why this approach is better than spatial-domain methods.

- Periodic noise appears as distinct frequency peaks in the Fourier spectrum.

- The notch filter removes only the unwanted frequencies.

- It preserves important image details and edges.

(e) Logical sequence of the solution.

- Capture the noisy image.

- Convert it into the frequency domain using FFT.

- Locate the periodic noise frequencies.

- Apply a notch reject filter.

- Perform IFFT to reconstruct the image.

Autonomous Vehicle Navigation.

(a) Interpret the scenario and explain the challenge.

→ Heavy rain introduces noise into the camera image, making lane markings and road edges difficult to detect. The challenge is to remove noise while keeping important edges sharp for safe navigation.

(b) Importance of preserving edges while removing noise.

- Lane markings define the driving path.
- Road edges help detect road boundaries.
- Sharp edges improve object detection.

(c) Recommend a suitable filtering technique.
Use a Bilateral Filter.

→ The bilateral filter smooths noisy regions while preserving edges by considering:

- * spatial distance between pixels.
- * Intensity (brightness) differences.

(d) Justify why this filter preserves important details.

- Removes rain noise effectively.
- Preserves lane markings and road boundaries.
- Prevents edge blurring.

(e) Present the explanation systematically.

- (i) Capture the noisy road image.
- (ii) Apply the bilateral filter.
- (iii) Reduce rain-induced noise.
- (iv) Preserve lane markings and road edges.
- (v) Detect lanes and obstacles accurately.
- (vi) Enable safe vehicle navigation.